Hugo Marquerie 06/03/2025

Métrica inducida

Proposición 1 (Métrica inducida). *Sea V un espacio vectorial normado,*

$$\implies \forall x, y \in V : d(x, y) = \|x - y\|$$

define una métrica en V, llamada la métrica inducida por la norma.

***Demostración*:** Basta con comprobar que se satisfacen las propiedades de la definición. Sean $x, y, z \in V$, entonces:

(i) $d(x, y) = \|x - y\| \geqslant 0$ y $d(x, y) = 0 \iff \|x - y\| = 0 \iff x = y$.

(ii) $d(x, y) = \|x - y\| = \|(-1)(y - x)\| = |-1|\ \|y - x\| = \|y - x\| = d(y, x)$.

(iii) $d(x, z) = \|x - z\| = \|(x - y) + (y - z)\| \leqslant \|x - y\| + \|y - z\| = d(x, y) + d(y, z)$. ■

Referenciado en

- `esp-banach`
- `prop-desigualdad-triangular-inversa-norma`
- `teo-metrica-inducida-iff-homogeneidad-traslaciones`

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.